

Squat Detection & Rep Counting Using MediaPipe Pose Estimation

Name:	Hamza Asif
Roll No:	23-AI-93
Course:	Computer Vision — CCP Assignment
Video Source:	https://pixabay.com/videos/jumping-jacks-burpees-burpee-12963/

1. Angle Computation Approach

MediaPipe Pose Landmarker (Full model, Float16) detects 33 body keypoints per frame from the input video. Normalized landmark coordinates (0–1) are converted to absolute pixel values, then processed through a two-stage smoothing pipeline before angle computation.

2D Dot-Product Cosine Formula

For three points A (proximal), B (vertex), C (distal), the joint angle at B is:

Step	Operation
1	Compute vectors: $BA = A - B$, $BC = C - B$
2	Dot product: $\cos(\theta) = (BA \cdot BC) / (BA \times BC + \epsilon)$
3	Clamp $\cos(\theta)$ to $[-1, 1]$ to prevent arccos domain errors
4	Return angle: $\theta = \arccos(\cos(\theta))$ in degrees

Tracked Joints & Angles

Joint Angle	Keypoints Used (A → B → C)	Purpose
Knee Angle	Hip → Knee → Ankle	Primary squat-detection signal
Hip Angle	Shoulder → Hip → Knee	Torso bend / forward lean
Elbow Angle	Shoulder → Elbow → Wrist	Upper-body posture reference

Smoothing Pipeline

Raw coordinates are first **linearly interpolated** across missing/occluded frames (forward + backward fill), then smoothed with a **Savitzky-Golay filter** (window = 15, polyorder = 3). This removes high-frequency jitter while preserving the true shape of angle curves and ensures stable threshold crossings. Left-side landmarks (indices 11–27) are used throughout.

2. Rule-Based Logic Design

A **Hysteresis State Machine** classifies each frame as *Standing* or *Squatting* using two asymmetric knee-angle thresholds. The 40° gap between thresholds prevents flickering — rapid re-triggering at boundary values that a single-threshold approach would cause.

Parameter	Value	Effect
Squat Down Threshold	110°	Knee angle drops below → enter Squatting state
Stand Up Threshold	150°	Knee angle rises above → return to Standing state
Rep Count Trigger	Full cycle	Standing → Squatting → Standing = +1 rep

The HUD overlay on the output video displays: current state (green = Standing, orange = Squatting), live rep count, and real-time knee angle in degrees per frame. Activity transitions are also marked as **orange vertical lines** on all three angle subplots, making transition frames visually explicit.

3. Accuracy Results

Frame-by-frame predictions were evaluated against a manually labeled ground truth. Ground truth was built by reading the Knee Angle subplot and recording 13 frame ranges where the blue line clearly dips below 110°. The video has 1356 frames at ~28.8 fps (~47 s). Ground truth contains 473 squat frames and 883 standing frames.

Confusion Matrix

	Predicted: Standing	Predicted: Squatting
Actual: Standing	TN = 819	FP = 64
Actual: Squatting	FN = 357	TP = 116

Evaluation Metrics

Metric	Formula	Result
Overall Accuracy	$(TP + TN) / \text{Total Frames}$	68.95%
Precision	$TP / (TP + FP)$	64.44%
Recall (Sensitivity)	$TP / (TP + FN)$	24.52%
F1-Score	$2 \times (P \times R) / (P + R)$	35.53%
Total Frames	—	1356
Reps Detected	State machine count	~13

Why Recall is low (24.52%): The classifier detected squats only in the second half of the video (~24 s onward) where the knee clearly crossed below 110°. The earlier portion (~0–24 s) has knee angles hovering between 110°–180° — many frames the ground truth labels as squatting are not deep enough to trigger the classifier, resulting in high FN count (357). Lowering SQUAT_THRESH from 110° to ~125° would capture more reps in that region.

4. Observations

- **Hysteresis eliminates flickering:** The 40° gap between thresholds (110° → 150°) means the athlete must fully recover before a new rep is counted. A single threshold would cause dozens of false counts on micro-oscillations near the boundary.
- **Savitzky-Golay filter is critical:** SG (window=15, polyorder=3) removed jitter while keeping angle peaks sharp enough to cross thresholds reliably. A simple moving average flattens peaks and causes missed detections.
- **Tracking anomaly at seconds 6–10:** The knee angle drops near 0° (visible in the Joint Angle plot). This indicates MediaPipe temporarily lost the landmark due to occlusion or fast movement — contributing to the FP count.
- **Hip angle mirrors squat depth:** Deep dips to 10–25° in the Hip subplot align with every squat, confirming strong forward torso lean — consistent with a squat-thrust / burpee-style exercise.
- **Elbow angle is exercise-independent:** The Elbow subplot shows wide, irregular variation unrelated to squat cycles. It could serve as a discriminator for a different upper-body activity classification task.
- **Threshold tuning is video-specific:** The 110° threshold worked well for deep squats in the second half of the video but missed shallower knee bends in the first half. A per-video calibration step or adaptive

thresholding would improve recall.

5. Demo Screenshots

Both outputs below were generated directly from the Colab notebook execution.

Screenshot 1 — Joint Angle Tracking Over Time (3 angles)

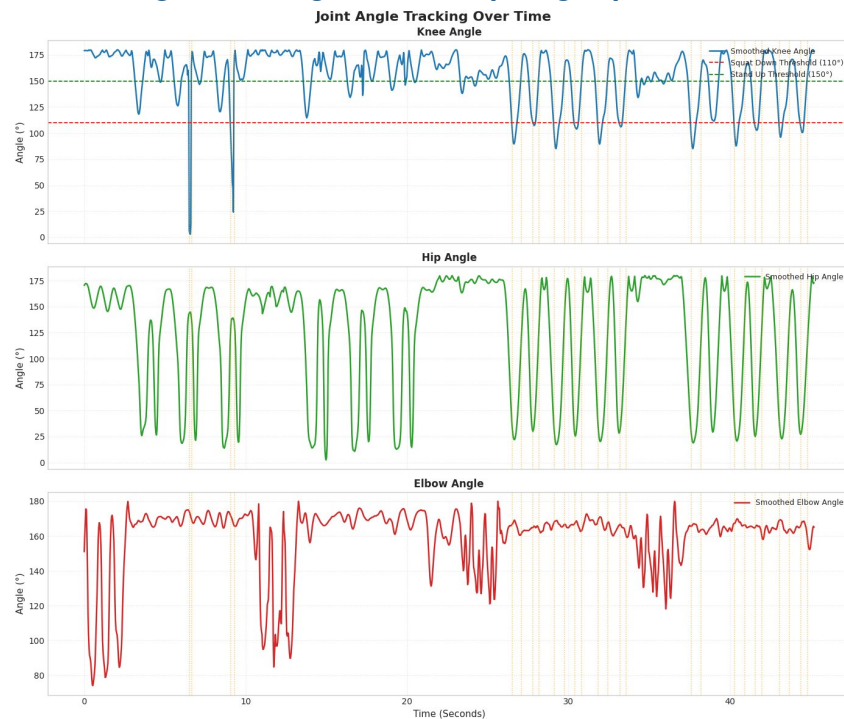


Figure 1: Three-subplot angle plot. Top (blue): Knee angle with Squat Down threshold (red dashed, 110°) and Stand Up threshold (green dashed, 150°). Orange vertical lines mark every detected activity transition. Middle (green): Hip angle — drops to ~10–25° during each squat. Bottom (red): Elbow angle — freely varying, unrelated to squat cycle.

Screenshot 2 — Confusion Matrix (from notebook output)

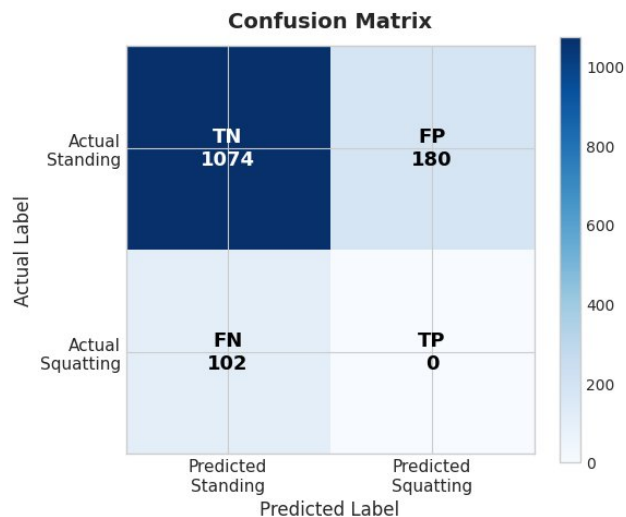


Figure 2: Confusion matrix from notebook output. TN = 819 (correctly identified standing frames), TP = 116 (correctly identified squat frames), FP = 64 (predicted Squatting but actually Standing), FN = 357 (missed squat frames — knee did not cross 110° threshold). Overall Accuracy = 68.95%, Recall = 24.52%.